Bias-to-Text:

Discovering and Mitigating Visual Biases through Keyword Explanation

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TL;DR:
We interpret visual biases through "keywords," which enables the discovery of novel biases and the effective debiasing of vision models.

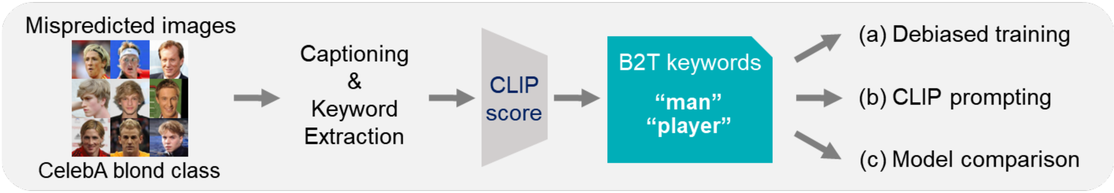
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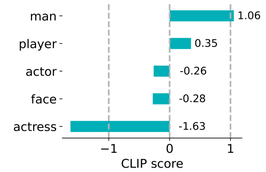
B2T (Bias-to-Text)

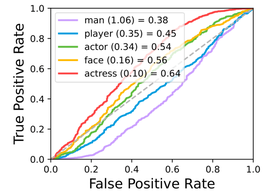
- Biases in models pose a critical issue when deploying AI systems into real-world but **diagnosing them in an explainable manner** can be challenging.
- To tackle this, we propose **B2T (Bias-to-Text)** that interprets visual biases as **"keywords"** by generating language descriptions directly from mispredicted images.
- Specifically, we first **extract B2T keywords with verification of CLIP score**, then use them for **various applications**.

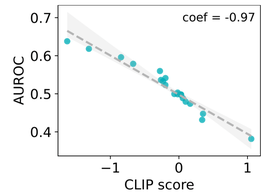
Step 1. Bias keywords generation Step 2. Various applications of keywords



CLIP score?

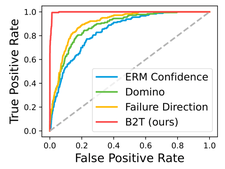






Bias discovery

- We discover ...
  - ✓ **spurious correlations**: (a), (b)
  - ✓ **distribution shifts**: (c), (d)
  - ✓ **unknown biases**: (e), (f)



... with best sample-wise bias labeling!

	(a) CelebA blond		(b) Waterbirds		(c) ImageNet-R		(d) ImageNet-C snow / frost	
Keyword	Man		Forest	Ocean	Illustration	Drawing	Snow	Window
Samples								
Actual	blond	blond	waterbird	landbird	backpack	white shark	airliner	American egret
Pred.	not blond	not blond	landbird	waterbird	maze	envelope	damselfly	quill
Caption	person, a **man** with a beard.	actor as a young **man**.	a bird in the **forest**.	a bird in the **ocean**.	hand drawn **illustration** of a backpack.	a **drawing** of a shark attacking [...]	airliner in the **snow**, photo.	a bird on a frozen **window**.

	(e) Dollar Street				(f) ImageNet			
Keyword	Cave	Fire	Bucket	Hole	Flower	Playground	Baby	Interior
Samples								
Actual	wardrobe	stove	plate rack	toilet seat	ant	horizontal bar	stethoscope	monastery
Pred.	poncho	caldron	oil filter	wheelbarrow	bee	swing	baby pacifier	arched ceiling
Caption	the **cave** is full of surprises.	a **fire** in the kitchen.	a **bucket** of water and a few tools.	the **hole** in the ground.	a yellow **flower** with a black head.	person on a swing in the **playground**.	a newborn **baby** boy in a stethoscope.	the **interior** of the church.

Applications

Debiased training

Method	GT	CelebA blond Worst	CelebA blond Avg.	Waterbirds Worst	Waterbirds Avg.
ERM	-	47.7±2.1	94.9	62.6±0.3	97.3
LfF [55]	-	77.2	85.1	78.0	91.2
GEORGE [74]	-	54.9±1.9	94.6	76.2±2.0	95.7
JTT [44]	-	81.5±1.7	88.1	83.8±1.2	89.3
CNC [86]	-	88.8±0.9	89.9	88.5±0.3	90.9
DRO-B2T (ours)	-	**90.4**±0.9	93.2	**90.7**±0.3	92.1
DRO [66]	✓	90.0±1.5	93.3	89.9±1.3	91.5

CLIP prompting

	CelebA blond Worst	CelebA blond Avg.	Waterbirds Worst	Waterbirds Avg.
CLIP zero-shot	76.2	85.2	50.3	72.7
+ Group prompt [85]	76.7	87.0	53.7	78.0
+ B2T-neg prompt	72.9	88.0	45.4	70.8
+ B2T-pos prompt (ours)	**80.0**	87.2	**61.7**	76.9

Label diagnosis

Keyword	Bee	Boar	Desk	Market
Samples				
Label	fly	pig	computer mouse	custard apple
Pred.	bee	wild boar	desktop computer	grocery store
Caption	a **bee** on a yellow flower.	wild **boar** in the forest.	the **desk** in the office.	fruit and vegetables at the **market**.

Model comparison

: ResNet vs. ViT

Keyword	Work		Supermarket	
Samples				
ViT-B	O	O	O	O
RN50	O	X	O	X
Actual (RN50)	dumbbell	dumbbell	shopping basket	shopping basket
Pred (RN50)	dumbbell	horizontal bar	shopping basket	grocery store
Caption	a set of dumbbells with weights.	person **works** out in the gym.	a basket full of food.	woman shopping in a **supermarket**.

Model comparison

: ERM vs. DRO

	Keyword	ERM	DRO	Gap
CelebA blond	man	1.06	✗	✗
Waterbird	bamboo forest	3.61	✗	✗
	bamboo	2.85	✗	✗
	forest	2.27	1.97	-0.30
	woods	2.24	1.88	-0.36
Landbird	seagull	3.10	1.85	-1.24
	beach	2.45	1.15	-1.30
	water	1.51	0.67	-0.84
	lake	1.25	✗	✗